

Cardiac Field Artifact Removal in Ear-EEG Using Repetitive Disturbance Rejection

Wenlin Zhang¹, Sam Lin^{1,6}, Shuai Li³, Ye Yuan¹,
Zheng Wang^{1,2}, Ziteng Ni⁴, Heng Kang Yao^{2,4}, Yongcheng Li⁵, Davide Momi⁷

Abstract—Continuous wearable EEG is increasingly paired with consumer wearables that provide a heartbeat stream (e.g., smartwatch ECG/PPG). Taking the available ECG/heartbeat reference, we propose a repetitive-control framework that models cardiac artifact as a heartbeat-aligned periodic disturbance. Each EEG sample is assigned a cardiac phase index wrapped around ECG R-peaks; phase-synchronous averaging then preserves heartbeat-correlated residue while attenuating non-phase-locked noise. Subtracting this phase-indexed average yields effective artifact removal in the non-causal offline setting. For online real-time operation, an iterative-learning estimator is used to track the residue in either a same-beat predictive mode (strict causal) or a one-beat-delayed mode (buffered causal). We also propose a PI based phase tracker for robust heartbeat timing estimation from raw EKG waveforms to account for breathe related heart rate variability or immature beats rejection. Results on open datasets and semi-synthetic simulations show that the heartbeat-timing-anchored repetitive/iterative learning approach provides strong artifact suppression at low compute cost, making it well suited to always-on wearable execution, where blind source separation is typically channel-starved.

I. INTRODUCTION

Wearable EEG is moving toward continuous, day-scale monitoring, and many users already wear a second device that provides heartbeat timing from ECG or PPG. This creates a practical path for low-channel ear-EEG denoising: use heartbeat timing already available from the wearable ecosystem rather than requiring additional EEG channels. In ear-centered montages, cardiac field artifact (CFA) is often prominent because electrodes are close to neck and vascular pathways. CFA is the volume-conducted electrical field of the heart [1], distinct from the scanner-related ballistocardiogram artifact emphasized in EEG-fMRI work [2]. For out-of-scanner earables, CFA is therefore a primary contamination mechanism.

Classical ECG-referenced adaptive cancellation is well established [3], and it is a natural baseline. However, in our target regime it faces two practical limits: earables are channel-starved for blind source separation [6], and reference-to-artifact waveform predictability can be weak under real volume-conduction mismatch and nonstationarity. Our real-data observations in this paper support exactly this point: coherence between ECG and ear-EEG artifact is close to

zero even with beat-synchronous pooling, so reference-driven waveform regression is limited as a primary mechanism. At the same time, beat timing remains highly informative, because the artifact is still phase-locked to R-peaks.

This motivates a timing-only control view: model cardiac contamination as a repetitive disturbance and reject it by heartbeat phase indexing. Repetitive control and iterative learning control (ILC) are mature tools for periodic disturbance rejection in mechatronics [8], and they map naturally to the heartbeat cycle when robust R-peak timing is available. We therefore use a min/max envelope tracker with PI/PID phase tracking and arrhythmia-aware supervision to produce stable beat timing, then apply phase-aligned template learning/cancellation in three deployment modes: non-causal offline, buffered causal (one-beat latency), and strict real-time causal (zero added latency).

The resulting contribution is a compact, wearable-oriented CFA removal scheme aligned with realistic device constraints: low channel count, low compute budget, and timing references that may come from a separate wearable. We evaluate the design on open real recordings and semi-synthetic data, compare against linear RLS-based and ICA baselines, and quantify the latency-versus-suppression tradeoff between non-causal, buffered, and strict-causal operation.

II. PROBLEM FORMULATION

Following repetitive-control modeling conventions [8], [9], we model the measured ear-EEG signal as

$$y(t) = s(t) + d_r(t) + n(t), \quad (1)$$

where $s(t)$ is the neural signal of interest, $d_r(t)$ is the heartbeat-synchronous repetitive cardiac disturbance, and $n(t)$ aggregates asynchronous interference and sensor noise. The subscript r denotes the repetitive nature of the disturbance. Its period is heartbeat dependent and is set by ECG R-peak timing, with beat interval

$$T_k = t_{k+1} - t_k, \quad (2)$$

where t_k is the k -th detected R-peak instant.

Assumption 1. *In ear-EEG, CFA is dominated by the repetitive component $d_r(t)$. The R-peak phase-asynchronous disturbances and noise $n(t)$ are assumed zero-mean across the R-peak sync averaging, and DC offset and slow drift are removed by preprocessing.*

Affiliations: ¹Institute of Brain and Mind, CA, USA; ²Krembil Centre for Neuroinformatics, Centre for Addiction and Mental Health, Toronto, Ontario, Canada; ³University of Oulu, Oulu, Finland; ⁴Synchroni Scientific Instruments Inc., Canada; ⁵Shenzhen Institutes of Advanced Technology, China; ⁶Palos Verdes Peninsula High School, CA, USA; ⁷University of California, Irvine, CA, USA; Corresponding author: Wenlin Zhang (e-mail: william@ibmind.org).

Assumption 2. The neural signal of interest $s(t)$ is not phase-locked to the R-peak: across beats, its phase relative to the cardiac cycle is effectively uniform. Consequently $s(t)$ is preserved by the proposed phase-synchronous repetitive runout cancellation, more detailed analysis is provided in Sec. III.

Using heartbeat phase, we represent the repetitive disturbance as a phase-indexed waveform

$$d_r(t) = d_r(\phi(t)), \quad \phi(t) = \frac{t - t_k}{T_k}, \quad t \in [t_k, t_{k+1}), \quad (3)$$

with $\phi(t) \in [0, 1)$ the R-peak-aligned cardiac phase.

Assumption 3. The beat period T_k is assumed to be accurately locked, and slowly varying due to respiratory sinus arrhythmia (RSA), with additional small modulation from low-level muscle/motion effects in resting-state recordings.

Problem 1. Find the repetitive cardiac artifact template $\hat{d}_r(\phi(t))$ such that subtracting it from the measured signal recovers the neural signal of interest:

$$\hat{s}(t) = y(t) - \hat{d}_r(\phi(t)) \approx s(t). \quad (4)$$

Under Assumptions 1–3, $\hat{d}_r(\phi(t))$ can be estimated from the R-peak-locked structure of $y(t)$ alone.

Note 1. Since $d_r(\phi(t))$ is periodic in the cardiac phase $\phi(t)$ indexed solely by R-peak instants $\{t_k\}$, Problem 1 depends only on R-peak timing and never on the ECG waveform shape. Consequently, any wearable heartbeat source—smartwatch ECG, wrist PPG, or any device delivering beat timing via a synchronisation mechanism—is sufficient to instantiate the solution.

Note 2. In reality, $s(t)$ may contain a phase-synchronous part that has non-zero expectation mean value across beats, e.g. the heartbeat-evoked potential (HEP)—a genuine cortical response time-locked to the cardiac cycle may or may not be absorbed into $\hat{d}_r$. In this paper, we make no claim that the solution of Eq. 4 preserves the phase-synchronous part of the neural signals, and it's a meaningful future work or extension worthy of further investigation.

To instantiate Problem 1 on the discrete sample grid, let $\hat{t}_k$ be the estimated k -th R-peak sample instant and f_s the sampling rate. The k -th beat's R-peak-aligned $d_r(\phi(t))$ at integer offset p is defined as

$$d_k\left(\hat{t}_k + \frac{p}{f_s}\right) \triangleq d_k(\phi_p), \quad (5)$$

with the discrete phase index and offset range

$$\phi_p \triangleq \phi\left(\hat{t}_k + \frac{p}{f_s}\right) = \frac{p}{f_s T_k}, \quad (6a)$$

$$p \in \{-n_{\text{pre}}, \dots, -1, 0, 1, \dots, n_{\text{post}}\}. \quad (6b)$$

Note 3. The template window spans $n_t = n_{\text{pre}} + n_{\text{post}} + 1$ samples ($p = 0$ at the R-peak). A practical choice is $n_{\text{pre}} = \lfloor 0.05 f_s \rfloor$ (50 ms, covering QRS onset) and $n_{\text{post}} = \lfloor 0.55 f_s \rfloor$ (550 ms, covering the full QRS-T complex); at $f_s = 250$ Hz this gives $n_t = 151$ samples.

III. REPETITIVE DISTURBANCE REJECTION

A. Non-Causal Cancellation

For offline CFA removal, we could run non-causal averaging to obtain the solution of Problem 1. That is, for a fixed phase index p (offset from the R-peak), the template value is the average, across N_r beats, of the EEG sampled at the same offset p from each detected R-peak $\hat{t}_k$:

$$\hat{d}_r(p) = \frac{1}{N_r} \sum_{k=1}^{N_r} y_k(p), \quad y_k(p) \equiv y(\hat{t}_k + p/f_s). \quad (7)$$

Substituting the signal $y_k(p)$ by its definition from the discrete time version of Eq. (1), we obtain

$$\hat{d}_r(p) = \underbrace{d_r(p)}_{\text{same every beat}} + \frac{1}{N_r} \sum_{k=1}^{N_r} s_k(p) + \frac{1}{N_r} \sum_{k=1}^{N_r} n_k(p). \quad (8)$$

In Eq. (7), we use $\hat{d}_r(p)$ to denote $\hat{d}_r(\phi_p)$ for better readability, and so are the p -indexed notations used in the subsequent discussion.

Take the expectation of Eq. (8) over R-peaks, we obtain

$$\mathbb{E}[\hat{d}_r(p)] = d_r(p) + \mathbb{E}[s_k(p)] + \mathbb{E}[n_k(p)]. \quad (9)$$

The repetitive R-peak-locked disturbance template $d(p)$ is deterministic. Ideally, The non-R-peak-locked neural term $\mathbb{E}[s_k(p)]$, and noise term $\mathbb{E}[n_k(p)]$ will render to zero according to Assumption 1, therefore the Eq. (8) will offer us solution by design:

$$\hat{d}_r(p) \approx d_r(p), \quad (10)$$

The CFA removed EEG signal is approximated by

$$\hat{s}_k(p) \approx y_k(p) - \hat{d}_r(p), \quad (11)$$

Let the asynchronous part at offset p have per-sample variance

$$\text{Var}[s_k(p) + n_k(p)] = V(p). \quad (12)$$

The noise floor leakage in $\hat{d}_r(p)$ due to limited sample average is the sample-mean of these N_r terms. If the per-beat samples are *mutually uncorrelated*,

$$\text{Var}[\hat{d}_r(p)] = \frac{1}{N_r^2} \sum_{k=1}^{N_r} \text{Var}[s_k(p) + n_k(p)] = \frac{V(p)}{N_r}, \quad (13)$$

IF we consider realistically there are some neuro signal has slight correlation with the R-peaks, the leakage will have an inflation term as

$$\text{Var}[\hat{d}_r(p)] = \frac{V(p)}{N_r} (1 + \delta_{coh}), \quad (14)$$

where δ_{coh} represents the total correlation accumulation across the beats. However, as the we choose a reasonable large enough N_r , the leakage and inflation will becomes negligible, we will give more detailed analysis in the next sections.

B. Causal Cancellation

For streaming operation the batch average of Eq. (7) is replaced by its standard exponentially-weighted ILC recursion, updated once per learnable beat at each offset p :

$$\hat{d}_r^{(k)}(p) = \hat{d}_r^{(k-1)}(p) + a [y_k(p) - \hat{d}_r^{(k-1)}(p)], \quad (15)$$

with reasonable gain $a = [0.06]$ for fast stable enough learning and minimum noise floor leakage; the supervisor freezes the update on premature/missed beats. Cancellation subtracts the template at the current phase offset, $\hat{s}(n) = y(n) - \hat{d}_r(p(n))$, where the anchor sets the mode: *buffered* anchors $p(n)$ to the detected R-peak (one-beat latency, full R-spike removal, matches non-causal), while *strict causal* anchors to the PLL-predicted R (zero latency) with acceptable predicted-R peak jitter. Both modes are $O(1)$ per sample with $O(L_w)$ memory, which makes the proposed method favorable for real-time implementation.

IV. EXPERIMENTS

A. Real Ear-EEG Validation

R-peak locking and record selection. On real recordings the phase reference is not given; we recover it from the raw ECG with a min/max-envelope tracker feeding a PI-based phase-locked loop (Sec. IV), which supplies the fiducials $\hat{t}_k$ that anchor the template. Because a heartbeat-phase-locked canceller is meaningful only where a heartbeat phase can be reliably established (Assumption 3), we restrict the real-data analysis to records on which the tracker attains stable locking, quantified by a low supervisor-flagged-beat fraction ($< [X]\%$) and a statistically significant R-locking test (below). Of $[N]$ ECG-bearing records screened, $[M]$ met this criterion and are analyzed here; the remainder—dominated by irregular rhythm or atypical, low-amplitude ECG morphology—fall outside the scope of a timing-only method and are reported separately. Fig. [LOCK] shows a representative locked segment and the non-causal template $\hat{d}_r(p)$ of Eq. (7).

Template SNR via a phase-scramble null. The ground-truth disturbance d_r is unobservable, so we validate the template against a null that shares its statistics but destroys R-phase locking. Define the R-locked energy of a signal z under a peak set P ,

$$\mathcal{E}_P[z] = \left\| \frac{1}{|P|} \sum_{k \in P} z(t_k + p/f_s) \right\|_2^2, \quad (16)$$

summed over the template window. Averaging on the true peaks gives $\hat{d}_r = d_r + \ell$, with leakage ℓ of scale $\text{SD}/\sqrt{N_r}$ (Eq. (13)); a circular time-shift of the peak train, which preserves the RR-interval sequence but destroys absolute phase, gives leakage only. Their ratio,

$$Q = \frac{\mathcal{E}_{\text{true}}[y]}{\mathcal{E}_{\text{scr}}[y]}, \quad \text{SNR}_{\text{templ}} = 10 \log_{10} Q, \quad (17)$$

is the CFA energy standing above the RRO-removed noise floor, with the leakage cancelling between numerator and denominator; a bias-corrected artifact energy follows as $\mathcal{E}_{\text{true}} -$

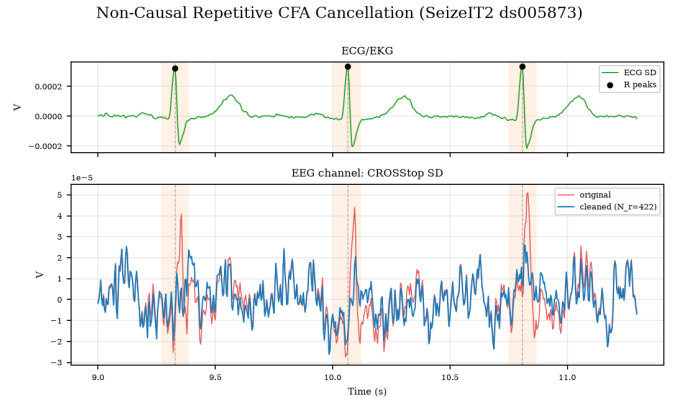


Fig. 1. Non-causal repetitive CFA cancellation on a short SeizeIT2 segment (ds005873, sub-070, ses-01, run-20; EEG index 1). Top: chest ECG with detected R-peaks (green). Bottom: original (red) and non-causally cleaned (blue) ear-EEG. The repetitive template $\hat{d}_r(p)$ by (Eq. (11)) is estimated from the full recording session ($N_r > 300$ beats).

$\overline{\mathcal{E}_{\text{scr}}}$. Repeating the scramble B times yields a null distribution and a p -value $p_Q = \frac{1}{B} \sum_b \mathbf{1}[\mathcal{E}_{\text{scr}}^{(b)} \geq \mathcal{E}_{\text{true}}]$. Across the selected records the template main lobe attains $\text{SNR}_{\text{templ}} = [XX]$ dB ($p < [0.01]$), confirming a genuine, well-estimated R-locked template rather than an averaging artifact. Fig. [SNR].

Coherence view. Template fidelity does not imply that the artifact waveform is predictable from the ECG. Fig. [COH] contrasts two coherences on the R-locked window: (i) between the ECG and the raw ear-EEG, diluted toward zero by the neural background ($\text{coh} \approx [0.1]$), and (ii) between the ECG and the high-SNR template $\hat{d}_r$ (non-causal and real-time), where dilution is removed and predictability can be tested honestly ($\text{coh} \approx [--]$, or FIR $R^2 \approx [--]$). This separates two distinct properties: the artifact is timing-repeatable ($Q \gg 1$) yet, on this electrode, only weakly waveform-predictable from the reference. That is precisely the regime in which a timing-only canceller is the appropriate tool and reference-driven waveform regression has little coherent structure to act on.

From a single operating point to a controlled sweep. The real recording fixes one low-predictability operating point and offers no ground-truth $s(t)$. To trace how reference-driven and timing-only cancellation behave as predictability varies, we turn to a semi-synthetic protocol (Sec. [X]) in which the ECG-to-artifact channel is swept from linear (high coherence) to strongly nonlinear (near-zero coherence) at fixed artifact power, quantifying the crossover at which reference regression collapses while the timing-only template holds.

Fig. 1 shows the result of non-causal repetitive disturbance rejection to the SeizeIT2 segment: the template is built from the full 5-minute recording ($N_r = 422$ learnable beats). Comparing the bottom panel of Fig. 1 with the original trace in the same panel, the heartbeat-locked deflections are substantially reduced.

Practically, as N_r is limited number, it's worthy to go deeper to look from variance perspective, about what residual leakage remains in the template.

The mean removes R-asynchronous components in expectation, but on a *finite* record a residual survives. Its size is a variance calculation.

Numerically: with $N_r = 359$ beats, ideal leakage is $1/\sqrt{359} \approx 5.3\%$ of the single-beat background standard deviation, which is around

TABLE I
STEADY-STATE ARTIFACT SNR (SEMI-SYNTHETIC; GENERATED).

Method	Latency	SNR (dB)
Input (contaminated)	–	1.27
Linear RLS ANC	causal	7.62
Proposed (causal)	0	8.60
Proposed (buffered)	1 beat	9.49
Proposed (non-causal)	offline	10.54

V. CONCLUSION

VI. CONCLUSION AND FUTURE WORK

We have presented a control-theoretic view of cardiac-field-artifact (CFA) removal in low-channel ear-EEG: CFA is modeled as a heartbeat-phase-locked repetitive disturbance and rejected by phase-indexed synchronous averaging and iterative learning, anchored only to R-peak timing and never to the ECG waveform. On real behind-the-ear recordings the ECG-to-artifact coherence is near zero even under best-case R-locked pooling, yet a clear, repeatable template is recovered—a phase-scramble null places it $[XX]$ dB above the RRO-removed noise floor. This makes the central claim precise: in this regime timing-only rejection is both *necessary* (reference-driven waveform regression has little coherent structure to act on) and *sufficient* (the artifact is genuinely timing-repeatable). A semi-synthetic sweep along the reference-quality axis locates the crossover: reference-driven cancellation (RLS/NRLS) is competitive only where the artifact is linearly predictable and collapses as coherence falls, whereas the timing-only template holds throughout—at $O(1)$ per-sample cost, $O(L_w)$ memory, and from a heartbeat source as light as a wrist PPG.

We are deliberate about where this matters. For a trained human electrographer a stereotyped, ~ 1 Hz contaminant is largely ignorable, and the modest broadband suppression it permits is of limited value. The benefit lies elsewhere—in automated, variance-sensitive readers, where a structured periodic bias corrupts an estimate out of proportion to its energy. Command decoders, continuous brain-state and depth-of-consciousness monitoring, and model-based neural inverse problems—for instance fitting Jansen–Rit / neural-mass observers to running EEG—all inherit a phase-locked artifact as a systematic bias in spectral, coherence, and connectivity features. Removing it at constant cost, without an additional electrode, is precisely the front end such wearable BCI pipelines require. In this sense the paper offers a *perspective* as much as a method: cardiac artifact is the clean, analyzable first instance of a reference-driven, control-based disturbance-observer subsystem for earable neural interfaces, and the

disturbance-observer framing gives a principled rule for when a reference is timing-only versus waveform-bearing.

Future work follows three directions. *Robust phase reference*: extending reliable R-peak and beat-phase estimation to arrhythmic and low-quality ECG, and to PPG-only timing, to lift the record-selection precondition of Sec. [X]. *The unifying observer*: the same reference-driven structure extends to an inertial (IMU) reference for motion artifact and an event-driven trigger for blink/jaw-clench transients, toward a single artifact-rejection subsystem, and to the high-CFA regime (e.g. neonatal or thin-skull montages) where suppression is large rather than modest. *Downstream validation*—most important for the benefit argued above—measuring, rather than assuming, the accuracy a variance-sensitive decoder or a neural-mass observer gains from CFA removal, together with an algorithm-unrolled, low-power on-device realization for always-on wearable execution.

More broadly, this work aims to bridge mature mechatronic disturbance-rejection theory and biosignal denoising, opening a control-systems avenue for wearable neural interfaces. The disturbance-observer perspective extends naturally beyond the cardiac artifact: an inertial (gyroscope/accelerometer) reference could drive feed-forward cancellation of motion artifact, and an event-driven detector could trigger transient cancellation of eye-blink or jaw-clench artifacts, suggesting a unified, reference-driven, control-based artifact-rejection subsystem for earable EEG. Future work includes validation on recorded ear-EEG, multichannel spatial coupling for $C \leq 8$, an RSA-aware R predictor to lift the zero-latency R-spike ceiling, and an algorithm-unrolled low-power realization.

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